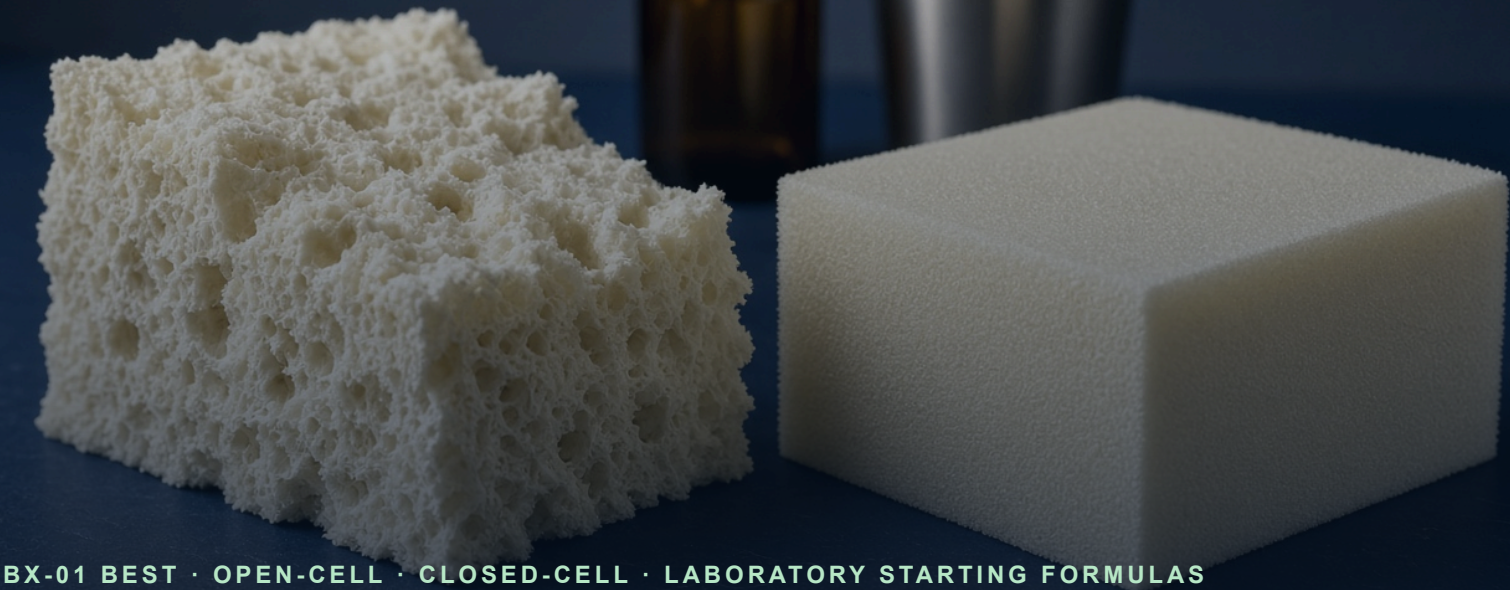


REV. J · FORMULATION PACK



BX-01 BEST · OPEN-CELL · CLOSED-CELL · LABORATORY STARTING FORMULAS

Open-Cell and Closed-Cell Foam

Formulas first. **BX-01** is the BazzLab card — the mix others do not publish. OC-01 and CC-01 stay as lab screens. TDS, catalyst, memo and patent notes follow so every number can be checked.

- 01** **Best formula · BX-01 / BX-02**
HFO-1233zd closed-cell on TB-307 / TEROL 250. Costs less than CREAPOL 326 / 1500. No Mannich. Unique to this floor.
- 01a** **50 g lab cup + checklist**
50 g CC (BX-01 + HFO) and 50 g OC (BX-02, no HFO), each with a tick-off bench list.
- 02** **Open-cell formula · OC-01 / OC-02**
High-water memo screen. Soft, air-open, 7–12 kg/m³.
- 03** **Closed-cell formula · CC-01 / CC-02**
Lower-cost swap starting card. Sealed cells, 28–40 kg/m³. Mix about 90:100 with PMDI.
- 04** **Polyol technical summary**
ISOEXTER TB-307 / KIMpol PE 104 and TEROL 250 / KIMpol PE 103 in the cards, with client TDS numbers. PD 230 stays.
- 05** **Catalysts and supplier memo ranges**
PO 1009433 catalyst roles and xu wang systems A / B / C.
- 06** **Mix method and trial matrix**
Cup procedure, index maths, and the first pours — BX-01 first. 50 g weigh sheet is a separate page.
- 07** **Patent notes and sign-off**
Honeywell / Arkema numbers from the client screenshots, plus use limits.

Best Formula · BX-01

BEST

NEW

This is the card to pour first. **HFO-1233zd(E) is mandatory** on closed cell. Memo System C and cup-test v4 are different recipes — do not blend them into this pour.

Why others do not have this

Typical spray copies use Mannich or sucrose for the network, CREAPOL 326 / 1500, and K-15 / DMCHA. They also use HFO — that part is now required here too.

BX-01 still does the rest differently. ISOEXTER TB-307 (f = 2.5) is the network — no Mannich. TEROL 250 (39.5 % aromatic) carries fire. PO 1009433 cats only. **HFO-1233zd(E) 10.0** is the blowing gas. Water is cut to 1.20.

Target

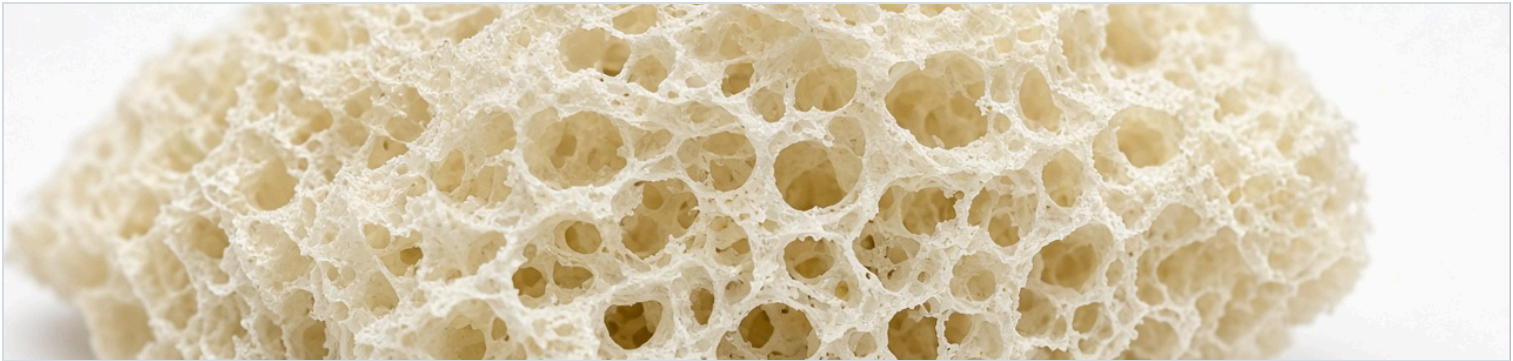
Density	32–38 kg/m³
Closed-cell content	> 90 %
Cream / gel / tack-free	3–6 s / 8–15 s / 15–28 s
Isocyanate	PMDI, 31.0 % NCO
Index	125 (calc.)
Weight ratio	≈ 83 A : 100 B

BX-01 · Closed-cell · HFO mandatory · 100.0 pbw B-side

Recommended first cup. Faster cream/gel pack. Confirm the drum label. Weigh HFO last, cold, then cap. Mix 4–5 s — cream is now 3–6 s.

INGREDIENT	WHY IT IS IN THIS CARD — NOT IN THE USUAL COPIES	PBW	OH#	EQ
ISOEXTER TB-307 / KIMpol PE 104	Lower-cost f = 2.5 rigid network. Replaces Mannich and CREAPOL 326.	41.3	305	0.2245
TEROL 250 / KIMpol PE 103	Lower-cost flow + 39.5 % aromatic fire. Replaces CREAPOL 1500 and extra TCPP.	24.8	260	0.1149
CREAPOL PD 230	Small toughness trim only. No lower-cost match.	4.0	225	0.0160
TCPP	Cut vs typical 12–23. Terol aromatic does part of the fire job.	6.0	0	0
TEP	Viscosity cut for thick PE 104 / TB-307.	4.0	0	0
Reactive FR	Bound phosphorus. Re-index if the grade has OH.	2.0	0*	0*
Emulsifier	Keeps 1.20 water in the thick polyester.	1.75	0	0
Closed-cell silicone	Sealed cells. Do not use an open-cell silicone.	1.5	0	0
Water	Cut because HFO does the blow. Do not go back to 1.75.	1.20	6233	0.1333
HFO-1233zd(E)	Mandatory closed-cell blowing gas. 0 OH. Weigh last, cold, cap.	10.0	0	0

Open-Cell Foam · OC-01



Open-cell structure. High water makes CO₂. Catalysts blow faster than they gel so the windows rupture.

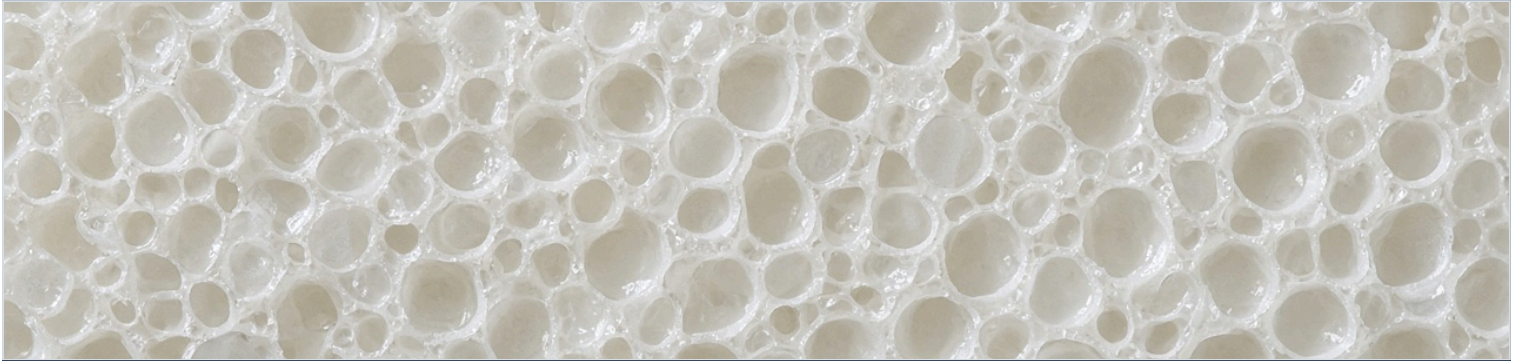
Target		
Density		7–12 kg/m³
Closed-cell content		< 10 %
Cream / rise / tack-free		2–5 s / 8–15 s / 10–20 s
Isocyanate		PMDI, 31.0 % NCO
Index, lab cup		90–105
Index, 1:1 spray		35–55

OC-01 · 100.0 pbw B-side

Midpoints of supplier System C. Amine package is 10.0 pbw: blow (PC-37 + CAT-T + BDMAEE) over gel (PC-15).

INGREDIENT	WHY IT IS IN THE FORMULA	PBW
Sucrose / sucrose polyol	High-functionality backbone. Builds a network that can open and still stand.	33.0
Reactive flame retardant	FR that binds into the polymer instead of only sitting in the blend.	8.0
TEP	Cuts viscosity and adds phosphorus FR.	17.0
Ecosurf SA-9	Emulsifies water into the polyol so blow is even.	6.0
Silicone surfactant	Opens and evens the cells. Too little = collapse. Too much = closed windows.	1.2
Water	Main blowing agent. Reacts with NCO to make CO ₂ . This is what makes the foam open and light.	24.0
PC-37	Blow catalyst.	2.4
CAT-T	Blow catalyst.	2.4
PC-15	Gel / balance. Holds the rise so it does not collapse.	4.2
BDMAEE or DEDEE	Strong open-cell blow. Helps rupture windows.	1.0
Biocide	In-can protection of the high-water blend.	0.8
B-side total		100.0

Closed-Cell Foam · CC-01



Closed-cell structure. Low water, higher index, and gel-heavy catalyst keep the membranes intact.

Target	
Density	28–40 kg/m³
Closed-cell content	> 90 %
Cream / gel / tack-free	8–15 s / 25–45 s / 40–70 s
Isocyanate	PMDI, 31.0 % NCO
Index	115–125 (calc. 120)
Weight ratio	≈ 90 A : 100 B

CC-01 · 100.0 pbw B-side

Water is 2.0 pbw, not 15–26. That is the difference between a closed-cell rigid foam and an open-cell spray foam.

INGREDIENT	WHY IT IS IN THE FORMULA	PBW	OH#	EQ
ISOEXTER TB-307 / KIMpol PE 104	Lower-cost high-OH rigid backbone. Replaces CREAPOL 326 GP.	35.5	305	0.1930
TEROL 250 / KIMpol PE 103	Lower-cost flow polyester. Replaces CREAPOL 1500. Carries pentane in CC-02.	24.0	260	0.1112
CREAPOL PD 230	Softer glycol polyester. Stops the blend going too brittle.	7.0	225	0.0281
TCP	Halogen FR.	12.0	0	0
TEP or DEHMP	FR plus viscosity cut.	6.5	0	0
Reactive FR	Bound FR. Re-index if it has OH groups.	5.0	0*	0*
Emulsifier	Keeps water and FR mixed.	3.5	0	0
Silicone surfactant	Fine, sealed cells. Do not use an open-cell silicone here.	1.4	0	0
Water	Limited CO ₂ . Enough to help nucleation, not enough to open cells.	2.0	6233	0.2222
PC-15	Gel / balance.	1.30	—	—
DABCO 2040	Strong gel / cure.	1.00	—	—
CAT-T	Small blow trim only.	0.25	—	—

Polyol Technical Summary

CC-01 and OC-02 use TB-307 / TEROL 250, which cost less than CREAPOL 326 / 1500. Mid-range OH figures are used only for the CC-01 index. Specification windows stay as published.

PROPERTY	TB-307 / PE 104	TEROL 250 / PE 103	CREAPOL PD 230
Replaces	CREAPOL 326 GP	CREAPOL 1500	Kept — no lower-cost match
Manufacturer NEW	COIM ISOEXTER TB-307 · Kimpur KIMpol PE 104	Huntsman TEROL 250 · Kimpur KIMpol PE 103	Flokser CREAPOL
Chemistry	Aromatic polyester, higher-OH rigid	Aromatic polyester for rigid PUR / PIR	Aromatic, glycol-based polyester polyol
Stated use	Spray, systems, bunstock. Fire resistance plus mechanical properties.	Spray, pour-in-place, panels, rigid foam	General aromatic polyester modifier
Hydroxyl number	295–315 mg KOH/g (both)	PE 103: 250–270 TEROL 250: 235–255	210–240 mg KOH/g
Mid OH used in calc.	305 mg KOH/g	260 mg KOH/g (PE 103 mid)	225 mg KOH/g
Equivalent weight	183.9 g/eq @ 305	Card: 215.8 @ 260 TEROL published: 224.4	249.3 g/eq
Viscosity @ 25 °C	TB-307: 7 000 cP PE 104: 7 500–8 500 mPa·s	PE 103: 4 000–5 000 mPa·s TEROL: 4 000–6 000 cP	4250–4750 cP
Functionality NEW	TB-307: 2.5 · PE 104: not published	TEROL 250: 2.0 · PE 103: high-functionality	—
Density / SG NEW	TB-307: 1.20 g/cm³	TEROL 250: 1.234 @ 25 °C	—
Acid number NEW	TB-307: max 1.50 mg KOH/g	—	≤ 1 mg KOH/g
Aromatic content NEW	—	TEROL 250: 39.5 %	—
Storage / shelf NEW	Request official TDS	TEROL: 15–50 °C · 6 months	20–40 °C · 12 months
Role in the formulas	CC-01 rigid backbone	CC-01 flow / pentane carrier · OC-02	CC-01 modifier · OC-02 trial

Client TDS pack — four grades, not two drums **NEW**

Copied from the Sep 2026 technical-data summaries. TB-307 is not PE 104, and TEROL 250 is not PE 103. They are same-role, lower-cost grades. These PDFs are source-derived summaries, not manufacturer-issued original TDS.

Catalysts and Supplier Ranges

Roles from PO 1009433. Chemical names are the industry spellings (the phone photos contain OCR typos). Quantities are listed only so the lab knows which drums are on the floor.

CODE	CHEMICAL NAME	ROLE IN FORMULA	ON THE PO	PACK	HS
PC-15	N,N,N',N'-Tetramethyldipropylenetriamine	Gel / balance	26 236 lb	1 874 lb × 14 totes	2921
PC-37	2-[2-(Dimethylamino)ethoxy]ethanol	Blow	1 984 lb	1 984 lb × 1 tote	2922.19
CAT-T	Trimethyl hydroxyethyl ethylenediamine	Blow	9 370 lb	1 874 lb × 5 totes	2922.1

Memo rule — most open cell

Blow	CAT-T / BDMAEE / PC-37 / DEDEE
Gel / balance	PC-15
Strong gel / cure	DABCO 2040
Combined OC cat package	USD 5.15 / kg CIF Turkey port

Still required for the formulas

Polymeric MDI (≈ 31 % NCO), silicone surfactant, emulsifier (NP-9 or Ecosurf SA-9), flame retardant (TCPP / TEP / DEHMP / reactive FR), biocide, and — for OC-01 — sucrose / polyether polyols. For every closed-cell cup, **HFO-1233zd(E) is mandatory**.

Supplier memo — OC/CC ingredient windows (pbw)

From Memo Style.pdf, 31 August 2026, xu wang to george@bazzfoam.com. These are ranges, not a finished 100-part formula. High water (15–26) marks A / B / C as open-cell type. Closed-cell water is 2.0 in CC-01.

INGREDIENT	A	B	C
Sucrose / sucrose polyol	20–60	—	30–36
Polyether polyol 1	—	15–19	—
Polyether polyol 2	—	14–18	—
Reactive FR	—	—	6–10
DEHMP / organophosphate	—	13–17	—
TEP	—	—	15–19
TCPP	15–25	—	—
Amine catalysts	4–10	8–12	8–12
Emulsifier	5–15	8–12 NP-9	4–8 Ecosurf SA-9
Silicone surfactant	0.5–2	0.5–2	0.5–2

Mix Method and Trial Matrix



Part B is the formula above. Part A is polymeric MDI. Bring both to temperature, then mix before cream.

Cup method

- Condition A and B at **21–23 °C**.
- Weigh B, then weigh A, into dry cups.
- Pour A into B. Mix **4–5 s** at 2 500–3 000 rpm for both OC and CC — cream is now 2–6 s.
- Clock cream, gel, rise and tack-free from mix start.
- Do not touch after cream. Cut at 24 h.

First cup: 50.00 g B (includes 5.00 g HFO) + 41.40 g A on the [50 g lab](#) sheet. No machine spray until three clean cups.

Working equations

$$EW_{\text{polyol}} = 56\,100 / \text{OH\#} \cdot \text{eq} = \text{pbw} / \text{EW} \cdot EW_{\text{water}} = 9.0 \cdot EW_{\text{PMDI}} = 4\,200 / \%\text{NCO}$$

$$\text{Index} = (\text{NCO eq} / \text{OH eq}) \times 100 \cdot \text{A-side (g)} = \text{OH eq} \times (\text{Index} / 100) \times EW_{\text{PMDI}}$$

First pours — BX-01 first, then screens

TRIAL	CHANGE FROM CONTROL	WHAT YOU ARE ASKING
BX-01	Best card — HFO 10.0 · faster cat pack	Cream 3–6 s · gel 8–15 s · does it seal?
BX-01a	HFO 10.0 → 11.0 / water 1.20 → 1.10	Lighter / more blow
BX-01b	CAT-T +0.15 / PC-15 +0.15	Still faster if 3–6 / 8–15 is missed
BX-02	Best OC, only if sucrose is on the floor	Commercial 16–22 kg/m³ open cell without polyethers
OC-01	Memo screen	Does System C open at 7–12 kg/m³?
CC-01	Lower-cost swap 90:100 screen	Fallback if BX-01 is too tight

Record

Temperature, mix time, cream, gel, rise, tack-free, collapse, shrink, scorch, friability.

Measure

Patent Notes and Sign-off

Numbers read from the client screenshots. Listed as blowing-agent / foam technology context, not freedom-to-operate advice.

APPLICANT (AS SHOWN)	PUBLICATION / PATENT	DATE ON SECOND SCREENSHOT
Honeywell	US7655610 / US7655610B2	2024.10
Honeywell	US8618040 / US8618040B2	2023.11
Honeywell	US8962707 / US8962707B2	2023.10
Honeywell	US9884946	2025.4.6
Honeywell	US8053404B2	2024.4
Honeywell	US8623808B2	2024.4
Arkema	CN105001440A	—
Honeywell	EP2640800A2 · EP2737006A2 · EP2606086A2 · EP3567072A1	—
Honeywell	JP5504161B2 · JP6231496B2	—

Source files used

ISOEXTER TB-307, KIMpol PE 103 / PE 104, TEROL 250 TDS summaries · TB307_PE104_TEROL250_PE103_TDS_Summary_Pack.pdf · CREAPOL 326 / 1500 / PD 230 TDS (reference) · Memo Style.pdf (xu wang, 31 Aug 2026) · PO 1009433 catalyst photos · Honeywell / Arkema patent screenshots.

What this pack is not

Not a production specification, fire-test certificate, spray-machine card, or legal clearance. Write the exact silicone, FR and PMDI grades on the trial sheet when the drums are opened.

Next actions

STEP	ACTION	OWNER
1	Confirm sucrose / polyether, silicone, FR, PMDI, and which floor drum is open (TB-307 vs PE 104, TEROL 250 vs PE 103).	Purchasing
2	Run BX-01 first as a 200–300 g cup at 22 °C. Then OC-01 / CC-01 only as screens.	Lab
3	Change only one variable per trial.	Lab
4	Closed-cell cups must include HFO-1233zd(E). Hood on. Chill B. Weigh HFO last.	Lab / HSE
5	Return cream / rise / density numbers for the next revision.	Technical

Issue and sign-off